Project Lasers: Review & Strategic Roadmap

Comprehensive Technical Review & Roadmap Update

1. Executive Summary

Project Lasers delivers a self-consistent, multi-physics telecom diode laser simulator designed to work within PINN-driven electronic design automation (EDA) loops. We have successfully implemented a decoupled 2D-1D multiscale wave-optical coupling model. By separating variables in the Helmholtz wave equation, we solve the 2D transverse mode shapes and 1D longitudinal traveling wave intensity separately. The longitudinal escape power at the facets is fed back dynamically to calibrate the cavity loss constant of the 2D PhotonSolver. This walkthrough details the physical models, robustness validation sweeps, and the updated strategic roadmap.

2. Current Implementation and Verification

- CarriersSolver: Nodal FVM solver for drift-diffusion equations, incorporating SRH, spontaneous, and Auger recombination. The carrier Jacobian is scaled to 1.0e-26 to ensure stable CoDoSol convergence.
- PhotonSolver: 2D scalar diffusion-reaction solver representing spatial photon density distribution, with decay tuned dynamically by the 1D Longitudinal Solver.
- EMSolver: Solves the 2D transverse Helmholtz eigenvalue problem to compute waveguide mode profile E_2d(y,z) incorporating state-dependent thermal and carrier index perturbations.
- ThermoSolver: Lattice heat solver driven by Joule heating, recombination heat, and optical absorption.
- Coupled Loop: Self-consistent solver loop converging cleanly in 15 steady-state iterations.

3. Robust Components (Key Strengths)

- Temperature Solver Stabilization: Using a Steady State Relaxation Factor of 0.5 only for the ThermoSolver dampens the thermal feedback loop and prevents numerical thermal runaway under active lasing forward bias.
- File Lock Robustness in Build Scripts: Added a rename catch block (.old rename workaround) in PowerShell_build.ps1. This allows successful DLL updates even if target library files are locked in memory by active Windows processes.
- Dynamic Wavelength/Mode Resolution: Solvers resolve the mode profile E_2d(y,z) and group velocity v_g self-consistently, matching actual physical modal gain rather than using static parameters.

4. Vulnerabilities and Risks

- Nodal Helmholtz Formulation: Solving vector field components (Ex, Ey) using nodal basis functions is susceptible to spurious (non-physical) modes at material interfaces. Edge elements should be introduced.
- Parameter Sensitivity: Recombination coefficients are heavily parameterized, which requires robust neural network inverse modeling against experimental measurements.

5. Strategic Roadmap (What to Do Next)

1. Migrate to Curl-Conforming Edge Elements: Rewrite EMSolver to use H(curl) edge elements to correctly represent tangential electric field continuity across material boundaries and eliminate spurious modes.
2. Automate PINN Calibration: Integrate the simulation pipeline with neural network models (pinn/) to dynamically calibrate active region recombination factors against experimental P-I-V measurements.
3. Multi-Mode and Spectral Resolution: Extend EMSolver to resolve multiple waveguide modes and longitudinal frequencies to simulate spectral line-shape broadening and side-mode suppression ratios (SMSR).

Status: Project Review Approved

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Date: July 17, 2026